SAARTHAK KHANDELWAL

AI & SOFTWARE ENGINEER

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EDUCATION

2026

VELLORE INSTITUTE OF TECHNOLOGY, VELLORE

- B.Tech Computer Science Engineering
- GPA- 7.82/10

2022

GREENWOOD HIGH, BANGALORE

- ICSE/ISC | X - 98%, XII - 92%

SKILLS

Programming Languages:
- Python, Java, C, C++, PHP, JavaScript

Frameworks & Tools:
- FastAPI, SentenceTransformers, FAISS, OpenAI API, Gemini API, Anthropic API, Qdrant, LangChain, HuggingFace, AutoGluon, PennyLane, Qiskit, REST APIs

Cloud:
- Azure (basic), AWS (basic)

Software:
- VS Code, Blue J, Power BI, Unreal Engine, Unity

Languages:
- English (Fluent), Hindi (Fluent), German (Basic), French (Basic)

PROJECTS

Neuroevolutionary Procedural Level Generator (Personal Project) — 2026
- Built a genetic algorithm in Python/NumPy that evolves 2D platformer levels, using a from-scratch neural network trained to learn a fitness function in place of hand-coded rules; validated with property-based testing (Hypothesis).

Autonomous Driving Simulator (Personal Project) — 2025
- Built a browser-based simulator in vanilla JavaScript where a neural network (trained via genetic algorithm) learns to drive using ray-cast sensor inputs, rendered live via Canvas API.

ONLINE COURSES

- CS50's Introduction to AI with Python - HarvardX
- Complete Generative AI with LangChain & HuggingFace - Krish Naik, Udemy
- Machine Learning Bootcamp - Devtown
- Complete 2024 Web Development Bootcamp - Angela Yu, Udemy
- Vue.js Fast Crash Course - Edwin Diaz, Udemy
- Complete C# Unity Game Developer 2D - Rick Davidson, Udemy

PROFILE

4th-year CS Engineering student at VIT Vellore specialising in AI, machine learning, and emerging technologies, with research and industry experience spanning quantum NLP, edtech AI, and backend development. Prospective MSCS candidate at the University of Sydney.

WORK EXPERIENCE

Calibr.AI (Brainhive Labs Pvt Ltd) — 2025 - 2026

AI Development Intern

- Enhanced the LMS platform with AI-driven modules and optimized REST APIs, improving content delivery and website performance for STEM and child education programs.
- Built a RAG-based pipeline for automated learning content generation, delivering dynamic, curriculum-aligned material for STEM and digital-skills programs.
- Engineered an automated skill tagging system using mpnet base v2 with FAISS-indexed taxonomy search and MMR-based diversity reranking, plus embedding-similarity difficulty classification served via FastAPI.

Vesuvius R&D (VRIPL), Vizag, India — 2025 - 2026

Research Intern

- Conducted predictive analysis on refractory lab trial data under the guidance of Dr. Akshay Bauskar, developing ML models to estimate key material properties.
- Designed a complete AutoGluon-based workflow for data cleaning, feature processing, multi-target regression modeling, and evaluation of refractory material behavior.

Vellore Institute of Technology, Vellore, India — 2025

Research Intern

- Worked under the guidance of Dr. Sathiya Kumar C., Professor, SCOPE.
- Developed a hybrid quantum-classical Transformer combining XLM-RoBERTa with VOC and QRC attention layers using PennyLane and Qiskit.
- Achieved 87% (EN) and ~80% (DE/ES) accuracy on XGLUE, outperforming classical baselines with accuracy-efficiency trade-offs analyzed across Hybrid, VQC, QRC, models using PennyLane and Qiskit.
- Preprint DOI: 10.5281/zenodo.21106178

Infinite Computer Solutions, Bangalore, India — 2024

Software Intern (Turbify)

- Navigated and analyzed a large backend codebase while integrating Apple Pay and implementing encrypted, secure data transfer mechanisms.
- Developed knowledge of encryption protocols and resolved real-time payment integration issues, optimizing backend performance and security.

Yale University, CT, USA — 2023 - 2024

Research Intern

- Worked under the guidance of Prof. Amit Khandelwal, Professor of Global Affairs and Economics.
- Analysed the effects of the US-China Trade War using R.
- Performed regression analysis in Python.